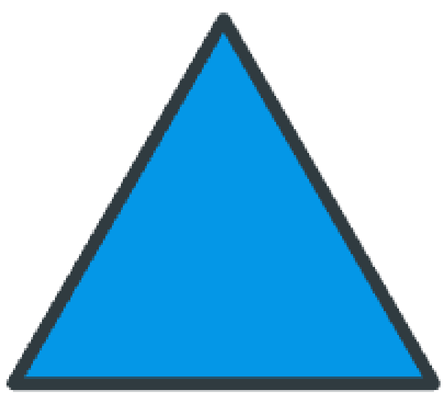
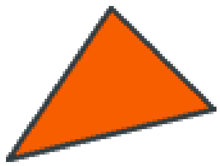
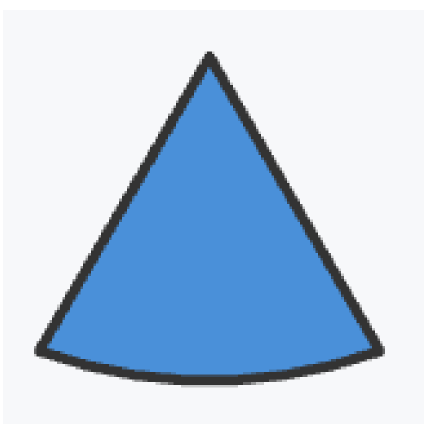
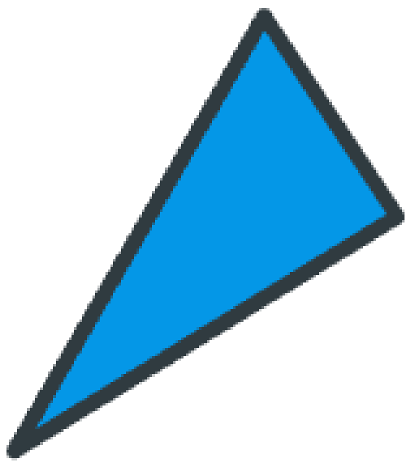
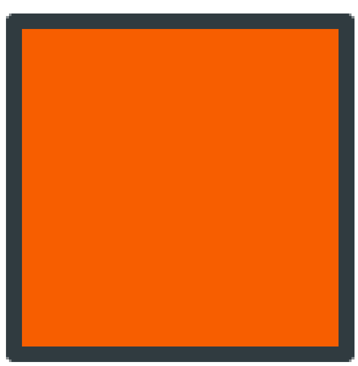
Lesson Objective

Explain decisions about classifications of triangles into categories using variants and non-examples. Identify shapes as triangles.

1. Identify a triangle by counting its straight sides and pointy corners. Is this shape a triangle? How do you know?  *Answer: Yes. It has 3 straight sides and 3 pointy corners.*	Show students the triangle. Point to each side one at a time and count aloud, "1, 2, 3." Have students whisper count along. Point to each corner one at a time and count, "1, 2, 3." Have students whisper count along. Tell students this shape is a triangle. A triangle has 3 straight sides and 3 pointy corners. Have students repeat the word triangle.
2. Decide whether each shape is a triangle by checking for three straight sides, three pointy corners, and a closed figure. Is each shape a triangle? How do you know? Shape A:	Show Shape A. Ask students to count the straight sides and the pointy corners. Ask if the shape is closed. Ask if it is a triangle and why. Show Shape B. Have students count the corners and sides. Point out that one side does not connect,

 Shape B:  Shape C:  *Answer: Shape A is a triangle (3 straight sides, 3 pointy corners, closed). Shape B is not a triangle (it is open). Shape C is not a triangle (one side is curved).*	so the shape is open. Ask if a pet inside this shape could get out. Ask if it is a triangle and why not. Show Shape C. Have students check each side. Point out that one side is curved, not straight. Ask if it is a triangle and why not. Remind students that a triangle must have all 3 things: 3 straight sides, 3 pointy corners, and a closed shape.
3. Circle the triangles and cross out the shapes that are not triangles.	Have students look at each shape on the worksheet.

<table>
<tr>
<td>Look at each shape. Circle it if it is a triangle. Cross it out if it is not a triangle.

Shape 1:

Shape 2:

Shape 3:</td>
<td>As students are ready, have them circle the triangles and cross out the shapes that are not triangles independently.

Check that students are counting sides and corners and looking for closed figures with all straight sides before deciding.</td>
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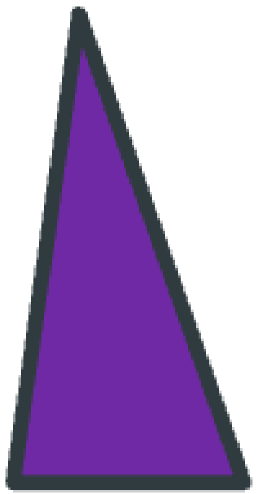
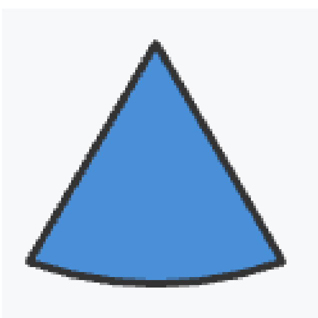
 Shape 4:  *Answer: Shape 1 circled (triangle); Shape 2 crossed out (4 sides, 4 corners); Shape 3 circled (triangle); Shape 4 crossed out (curved side).*	

Monitor For	**Instructional Moves**

• Do students count each side and corner with one-to-one correspondence, touching one and saying one number? • Do students count a corner as a side, or skip a side when counting? • Do students recognize the rotated triangle in Problem 3 as a triangle, even though it does not sit on a flat base? • Do students check all 3 attributes (straight sides, pointy corners, closed) in Problem 2, or decide based on how the shape looks?	• Model touching each side as you count, "1 side, 2 sides, 3 sides," so students see how to track sides one at a time without losing place. • If a student miscounts, guide them to start at one corner and trace the side all the way to the next corner before saying the next number. • For the open shape in Problem 2, ask, "If a bug were inside this shape, could it walk out?" to make the meaning of closed concrete. • If a student says Shape 1 in Problem 3 is not a triangle because it is "upside down," turn the worksheet so the shape sits flat and ask them to count again. Then turn it back and ask if it is still a triangle.
Reflection Questions • What 3 things do we look for in a triangle? *Answer: 3 straight sides, 3 pointy corners, and a closed shape.* • The shape in Problem 2 with the curved side had 3 corners. Why was it not a triangle? *Answer: One side was curved. Triangles have all straight sides.* • In Problem 3, one triangle was pointing a different way than the others. How did you know it was still a triangle? *Answer: It still had 3 straight sides, 3 pointy corners, and was closed.*	